

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	2 MARKS

5.2 – Sprint Delivery Plan

The *Sprint Delivery Plan* for the **Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study** follows an Agile methodology to ensure incremental development, continuous feedback, and timely delivery of project outcomes. Sprint 1 (Weeks 1–2) focuses on project initiation, requirement gathering, stakeholder analysis, and defining key performance indicators (KPIs). During this sprint, the team identifies data sources, establishes project objectives, and prepares the overall project roadmap. Sprint 2 (Weeks 3–4) is dedicated to data collection and integration, where student survey data, cafeteria transaction records, nutritional databases, and demographic information are gathered and consolidated into a centralized repository. Sprint 3 (Weeks 5–6) covers data cleaning, transformation, and ETL development, including handling missing values, removing duplicates, validating data quality, and preparing datasets for analysis. Sprint 4 (Weeks 7–8) focuses on data modeling and exploratory analysis, where relationships between dietary habits, spending behavior, and nutritional outcomes are established, and KPIs are calculated to generate meaningful insights.

Sprint 5 (Weeks 9–10) is dedicated to Tableau dashboard development, involving the design of interactive visualizations, filters, drill-down capabilities, and performance optimization to provide a user-friendly analytics experience. Sprint 6 (Weeks 11–12) concentrates on dietary strategy formulation, where insights from the dashboards are translated into actionable recommendations such as personalized meal plans, healthy menu enhancements, nutrition awareness campaigns, and budget-friendly food options. Sprint 7 (Weeks 13–14) focuses on

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testing, validation, and deployment, ensuring dashboard accuracy, usability, security, and successful implementation on Tableau Server or Tableau Cloud. Finally, Sprint 8 (Weeks 15–16) emphasizes monitoring, feedback collection, performance evaluation, and continuous improvement. Stakeholder reviews are conducted at the end of each sprint to assess progress, validate deliverables, and incorporate feedback into subsequent iterations. This sprint-based approach ensures efficient project execution, minimizes risks, enhances collaboration among stakeholders, and delivers a robust, data-driven solution that promotes healthier food choices and improved student well-being through actionable Tableau insights.

SPRINT DELIVERY PLAN

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 **Goal:** Leverage data and Tableau analytics to understand college students' food choices, promote healthy eating habits, and design effective dietary strategies. 

	SPRINT 1 (Week 1 – 2)	SPRINT 2 (Week 3 – 4)	SPRINT 3 (Week 5 – 6)	SPRINT 4 (Week 7 – 8)	SPRINT 5 (Week 9 – 10)	SPRINT 6 (Week 11 – 12)	SPRINT 7 (Week 13 – 14)	SPRINT 8 (Week 15 – 16)
Focus Area	 Project Initiation & Requirement Analysis	 Data Collection & Integration						
Objective	Define project scope, goals, stakeholders and KPIs.	Collect and integrate data from multiple sources.	Clean, validate and prepare data for analysis.	Build data models and analyze to uncover key insights.	Design and build interactive dashboards and reports.	Translate insights into actionable dietary strategies.	Test, validate and deploy dashboards and reports.	Monitor impact and continuously improve strategies.
Key Tasks	- Stakeholder meetings - Define objectives & scope - Identify KPIs - Research & data source identification - Project plan & roadmap	- Design surveys - Collect survey responses - Extract cafeteria transaction data - Import nutritional database - Integrate demographic data - Ensure data privacy	- Data profiling - Handle missing values - Remove duplicates - Standardize formats - Data validation & quality checks - Build ETL workflows	- Create data model - Define relationships - Calculate KPIs & metrics - Exploratory data analysis - Identify trends & patterns - Segment student groups	- Design dashboard wireframes - Build interactive dashboards - Create visualizations & KPIs - Add filters, parameters, drill-downs - User testing & optimization	- Identify nutritional gaps - Develop meal recommendations - Propose healthy menu improvements - Design awareness programs - Budget-friendly meal strategies	- Functional testing - Data validation - User acceptance testing - Fix issues & refinements - Deploy to Tableau Server/ Tableau Cloud - Documentation	- Monitor dashboard usage - Track KPIs & outcomes - Collect stakeholder feedback - Update data & dashboards - Refine strategies - Continuous improvement
Deliverables	 Project Charter Requirements Document KPI List, Project Roadmap	 Raw Datasets Data Source Inventory Integrated Data	 Cleaned Datasets ETL Workflows Data Dictionary	 Data Models KPIs & Metrics EDA Insights Report	 Tableau Dashboards Reports User Guide	 Strategy & Recommendation Report Meal Plan Templates Implementation Plan	 Validated Dashboards Deployment Report Documentation	 Performance Report Updated Dashboards Feedback Summary
Tools / Tech	 MS Teams, Confluence, Excel, Google Docs	 MySQL, APIs, Google Forms, CSV	 Python, Pandas, SQL, AWS Glue	 MySQL, SQL, Excel, DAX (Tableau Prep)	 Tableau Desktop, Tableau Prep	 Excel, PowerPoint, Tableau	 Tableau Server/Cloud, Postman, JMeter	 Tableau, Excel, Power BI (optional)
Duration	 2 Weeks	 2 Weeks	 2 Weeks	 2 Weeks	 2 Weeks	 2 Weeks	 2 Weeks	 2 Weeks
Sprint Goal	Clear understanding of goals, scope, KPIs and plan.	All relevant data collected and integrated.	High-quality, clean and analysis-ready data.	Key insights and KPIs identified.	Interactive dashboards ready for review.	Actionable strategies & recommendations defined.	Dashboards tested, validated and deployed.	Impact measured & strategies continuously improved.

